

Preliminary Notation

$$\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C} \rightarrow \mathbb{Q} = \left\{ \frac{a}{b} : a, b \in \mathbb{Z} \text{ and } b \neq 0 \right\}$$

$$\mathbb{C} = \{x+iy : x, y \in \mathbb{R}\}$$

$$(a, b) = \text{open interval from } a \text{ to } b = \{x \in \mathbb{R} : a < x < b\}$$

$$[a, b] = \text{closed interval from } a \text{ to } b = \{x \in \mathbb{R} : a \leq x \leq b\}$$

$$[a, \infty) = \{x \in \mathbb{R} : a \leq x\}$$

$$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R}$$

Absolute Value function

$$\text{Represented with } |x| : x \in \mathbb{R}, |x| = \begin{cases} -x, & x < 0 \\ x, & x \geq 0 \end{cases}$$

$|a-b|$ = distance between 2 real numbers

Points in a plane

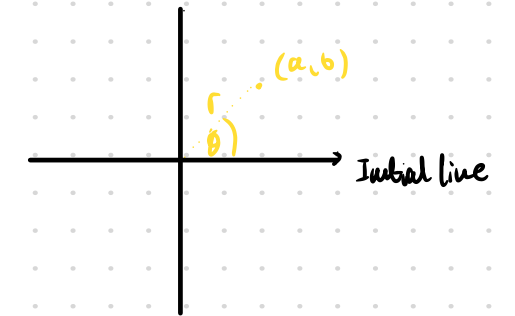
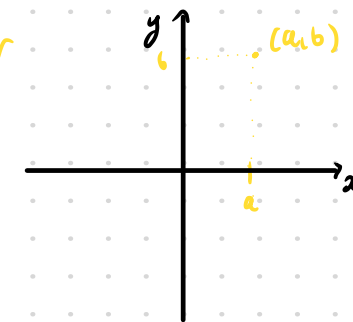
$$\text{Cartesian} \rightarrow (x, y) \in \mathbb{R}^2$$

$$\text{e.g. } \{(x, y) \in \mathbb{R}^2 : f(x, y) = 0\} \text{ where } f: \mathbb{R}^2 \rightarrow \mathbb{R} \text{ (mapping)}$$

$(0, 0)$ sometimes denoted as O

Conic Sections Introduction

Cartesian vs Polar



$$\text{Consider } (x, y) \in \mathbb{R}^2$$

$$\text{general conic section form} = Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0, (A, B, \dots, F \in \mathbb{R})$$

Special Cases

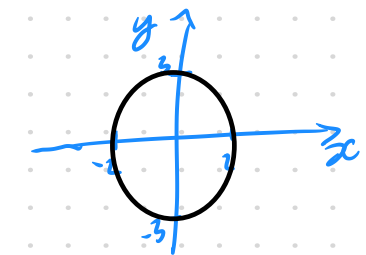
$$\text{Consider } (x, y) : (x-a)^2 + (y-b)^2 = r^2, a, b, r \in \mathbb{R}$$

$$\text{Circle Center} = (a, b) \text{ and radius / loci distance} = r \forall x, y \in \mathbb{R}$$

$$\text{Consider } (x, y) \in \mathbb{R}^2 : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 = \text{Ellipse}$$

Exercise

$$\frac{(x-1)^2}{2^2} + \frac{(y-2)^2}{3^2} = 1 \rightarrow$$



$$\text{Consider } (x, y) \in \mathbb{R}^2 : y^2 = 4ax, a \in \mathbb{R} = \text{Parabola}$$

